

C Programming - Error Analysis

Programs 1 to 5 | ReadMe Debugging Methodology

PROGRAM 1: Student Marks System

Step 1: Read the Entire Program

What is this program trying to do?

This program tries to manage student marks. It calculates total and average marks, stores scores, reads attendance, subject name, and prints various array values.

Identified Elements:

- Variables: total, average, present, totalMarks
- Arrays: marks[5], scores[5], attendance[5], arr[5], grade[5], prices[5], inventory[5], salary[5], result[5]
- Functions: main() only
- Loops: for loops for traversal and input
- Structures: None
- File Operations: None
- Pointers: None

Step 2: Syntax Errors

Line	Error Type	Description	Correction
6	Syntax	Missing semicolon at end of array declaration: <code>int marks[5] = {80,90,70,60,50}</code>	<code>int marks[5] = {80,90,70,60,50};</code>
86	Syntax	<code>grade[0] = "A"</code> - double quotes used for char, should be single quotes	<code>grade[0] = 'A';</code>

Step 3: Semantic Errors

Line	Error Type	Description	Correction
23	Semantic	Wrong format specifier: <code>printf("%d", average)</code> but average is float	<code>printf("Average = %f\n", average);</code>
84	Semantic	<code>scanf("%d", arr[i])</code> - missing & operator for address; should be <code>&arr[i]</code>	<code>scanf("%d", &arr[i]);</code>
86	Semantic	<code>grade[0] = "A"</code> - string literal assigned to char element	<code>grade[0] = 'A';</code>
95	Semantic	<code>if(prices[2] = 300)</code> - assignment used instead of comparison, always true	<code>if(prices[2] == 300)</code>
103	Semantic	<code>inventory = prices</code> - cannot copy array by assignment in C	<code>Use memcpy(inventory, prices, sizeof(prices));</code>
110	Semantic	<code>printf("%d", salary[0])</code> but salary[0] is float - wrong format specifier	<code>printf("%f\n", salary[0]);</code>
117	Semantic	<code>city = "Jaipur"</code> - cannot assign string to char array after declaration	<code>strcpy(city, "Jaipur");</code>

Step 4: Scope Errors

None found in this program.

Step 5: Linker Errors

None found in this program.

Step 6: Runtime Errors

Line	Error Type	Description	Correction
12	Runtime	for(i=0; i<=5; i++) - loop runs 6 times, marks[5] is out of bounds for marks[5]	for(int i=0; i<5; i++)
33	Runtime	scores[5] = 60 - index 5 is out of bounds for int scores[5] (valid: 0-4)	Remove scores[5] or declare scores[6]
45	Runtime	strcpy(subject, "Programming") - 11 chars into char subject[5], buffer overflow	char subject[15]; strcpy(subject, "Programming");
53	Runtime	attendance["one"] = 1 - string used as array index, invalid	attendance[0] = 1; (use integer index)
55	Runtime	attendance[2.5] = 1 - float used as array index, invalid	attendance[2] = 1; (use integer index)
61	Runtime	for(i=0; i<=5; i++) accessing attendance[5] - out of bounds for attendance[5]	for(int i=0; i<5; i++)
126	Runtime	for(i=0; i<=5; i++) accessing result[5] - out of bounds for result[5]	for(int i=0; i<5; i++)

Step 7: Logical Errors

Line	Error Type	Description	Correction
21	Logical	average = total / 5 - integer division loses decimal part (e.g. 350/5 should be 70.0 but gives 70 as int then stored in float)	average = total / 5.0;
95	Logical	if(prices[2] = 300) - assignment not comparison; condition is always true regardless of actual value	if(prices[2] == 300)

Step 8: Check Arrays Carefully

Line	Error Type	Description	Correction
12	Array	Loop i<=5 on marks[5] - accesses marks[5] which is out of bounds	Change to i<5
33	Array	scores[5] = 60 - out of bounds write for int scores[5]	Remove or resize to scores[6]
45	Array	char subject[5] too small for strcpy of 11-char string "Programming"	char subject[15]
53	Array	attendance["one"] - string index invalid	Use integer index
55	Array	attendance[2.5] - float index invalid	Use integer index

Line	Error Type	Description	Correction
61	Array	Loop i<=5 on attendance[5] - out of bounds	Change to i<5
103	Array	inventory = prices - array-to-array assignment not allowed in C	Use memcpy()
126	Array	Loop i<=5 on result[5] - out of bounds at result[5]	Change to i<5

Step 9: Check Functions Carefully

None found in this program.

Step 10: Check Structures

None found in this program.

Debugging Checklist - Program 1

- ☒ Syntax Errors
- ☐ Undeclared Variables
- ☒ Type Mismatches
- ☐ Function Call Errors
- ☐ Scope Errors
- ☐ Shadowing Issues
- ☐ Linker Errors
- ☒ Array Bounds Errors
- ☐ Pointer Errors
- ☐ File Handling Errors
- ☒ Runtime Errors
- ☒ Logical Errors
- ☒ Wrong Formulas
- ☒ Wrong Conditions

PROGRAM 2: Array Operations Program

Step 1: Read the Entire Program

What is this program trying to do?

This program performs various array operations. It fills a numbers array, finds sum and average, finds highest marks, tracks attendance, handles string arrays (subject, branch, college names), manages fees, stock, salary, inventory and city data.

Identified Elements:

- Variables: sum, average, highest, present, totalResult
- Arrays: numbers[10], marks[5], attendance[5], subject[10], branch[20], college[10], fees[5], result[5], stock[5], backup[5], salary[5], inventory[5], city[8], count[5]
- Functions: main() only
- Loops: multiple for loops for filling and printing arrays
- Structures: None
- File Operations: None
- Pointers: None

Step 2: Syntax Errors

Line	Error Type	Description	Correction
155	Syntax	Missing semicolon: printf("Program Ended\n")	printf("Program Ended\n");

Step 3: Semantic Errors

Line	Error Type	Description	Correction
35	Semantic	printf("Average = %d", average) but average is float - wrong format specifier	printf("Average = %f\n", average);
72	Semantic	scanf("%d", result[i]) - missing & for address of variable	scanf("%d", &result[i]);
81	Semantic	if(stock[2] = 300) - assignment instead of comparison	if(stock[2] == 300)
87	Semantic	backup = stock - cannot copy array by direct assignment in C	Use memcpy(backup, stock, sizeof(stock));
94	Semantic	printf("%d", salary[0]) but salary[0] is float - wrong format specifier	printf("%f\n", salary[0]);
61	Semantic	branch = "IT" - cannot assign string to char array after declaration	strcpy(branch, "IT");
99	Semantic	count = inventory - cannot copy array by direct assignment in C	Use memcpy(count, inventory, sizeof(inventory));

Step 4: Scope Errors

None found in this program.

Step 5: Linker Errors

None found in this program.

Step 6: Runtime Errors

Line	Error Type	Description	Correction
17	Runtime	for(i=0; i<=10; i++) on numbers[10] - accesses numbers[10] which is out of bounds	for(int i=0; i<10; i++)
27	Runtime	sum = numbers[i] inside loop - this is assignment not accumulation; overwrites sum each iteration	sum += numbers[i];
56	Runtime	attendance[5] = 1 - index 5 is out of bounds for attendance[5] (valid: 0-4)	Remove or declare attendance[6]
60	Runtime	for(i=0; i<=5; i++) on attendance[5] - accesses attendance[5] out of bounds	for(int i=0; i<5; i++)
66	Runtime	strcpy(subject, "ComputerScience") - 15 chars into char subject[10], buffer overflow	char subject[20]; strcpy(subject, "ComputerScience");
68	Runtime	fees["two"] = 50000 - string used as array index, invalid	fees[1] = 50000; (use integer index)
70	Runtime	fees[3.5] = 60000 - float used as array index, invalid	fees[3] = 60000; (use integer index)
97	Runtime	for(i=0; i<=5; i++) on inventory[5] - out of bounds at inventory[5]	for(int i=0; i<5; i++)
105	Runtime	strcpy(city, "JaipurCity") - 10 chars into char city[8], buffer overflow	char city[15]; strcpy(city, "JaipurCity");

Step 7: Logical Errors

Line	Error Type	Description	Correction
33	Logical	average = sum / 10 - integer division; sum is int so decimal part lost	average = sum / 10.0;
27	Logical	sum = numbers[i] overwrites sum with last element instead of adding all - wrong accumulation	Change to: sum += numbers[i];
46	Logical	Finding highest marks: if(marks[i] < highest) sets highest to smaller value - finds lowest, not highest	if(marks[i] > highest) to find actual highest
81	Logical	if(stock[2] = 300) assigns 300 to stock[2] instead of comparing - always true	if(stock[2] == 300)

Step 8: Check Arrays Carefully

Line	Error Type	Description	Correction
17	Array	Loop i<=10 on numbers[10] - out of bounds at index 10	Change to i<10
56	Array	attendance[5] = 1 - out of bounds for attendance[5]	Remove or use attendance[6]
60	Array	Loop i<=5 on attendance[5] - out of bounds	Change to i<5
66	Array	char subject[10] too small for 15-char string "ComputerScience"	char subject[20]
68	Array	fees["two"] - string used as index	Use integer index fees[1]
70	Array	fees[3.5] - float used as index	Use integer index fees[3]
87	Array	backup = stock - array assignment not allowed in C	Use memcpy()
97	Array	Loop i<=5 on inventory[5] - out of bounds	Change to i<5
99	Array	count = inventory - array assignment not allowed in C	Use memcpy()
105	Array	char city[8] too small for 10-char string "JaipurCity"	char city[15]

Step 9: Check Functions Carefully

None found in this program.

Step 10: Check Structures

None found in this program.

Debugging Checklist - Program 2

☒ Syntax Errors

☐ Undeclared Variables

☒ Type Mismatches

☐ Function Call Errors

☐ Scope Errors

☐ Shadowing Issues

☐ Linker Errors

☒ Array Bounds Errors

☐ Pointer Errors

☐ File Handling Errors

☒ Runtime Errors

☒ Logical Errors

☒ Wrong Formulas

☒ Wrong Conditions

PROGRAM 3: Student Result Analysis

Step 1: Read the Entire Program

What is this program trying to do?

This program analyzes student results. It calculates total, average, highest, and lowest marks, tracks attendance, reads branch name, manages fees, stock, backup arrays, salary, result input, city, inventory, college name, count, course name, and a numbers array.

Identified Elements:

- Variables: total, average, highest, lowest, present, totalResult
- Arrays: marks[5], attendance[5], subject[8], branch[10], fees[5], stock[5], backup[5], salary[5], result[5], city[10], inventory[5], college[5], count[5], course[6], numbers[5]
- Functions: main() only
- Loops: for loops for traversal and input
- Structures: None
- File Operations: None
- Pointers: None

Step 2: Syntax Errors

Line	Error Type	Description	Correction
165	Syntax	Missing semicolon: return 0	return 0;

Step 3: Semantic Errors

Line	Error Type	Description	Correction
78	Semantic	scanf("%s", &branch;) - & not needed for char array; array name already gives address	scanf("%s", branch);
105	Semantic	scanf("%d", result[i]) - missing & for address	scanf("%d", &result[i]);
112	Semantic	city = "Jaipur" - cannot assign string to char array after declaration	strcpy(city, "Jaipur");
123	Semantic	inventory["one"] = 100 - string used as array index, invalid	inventory[0] = 100;
125	Semantic	inventory[2.5] = 200 - float used as array index, invalid	inventory[2] = 200;
95	Semantic	backup = stock - cannot copy array by direct assignment in C	Use memcpy(backup, stock, sizeof(stock));
100	Semantic	printf("%d", salary[0]) but salary[0] is float	printf("%f\n", salary[0]);
139	Semantic	count = inventory - cannot copy array by direct assignment in C	Use memcpy(count, inventory, sizeof(inventory));
145	Semantic	strcpy(course, "Computer") - 8 chars into char course[6], buffer overflow	char course[10]; strcpy(course, "Computer");

Step 4: Scope Errors

None found in this program.

Step 5: Linker Errors

None found in this program.

Step 6: Runtime Errors

Line	Error Type	Description	Correction
60	Runtime	for(i=0; i<=5; i++) on attendance[5] - accesses attendance[5] out of bounds	for(int i=0; i<5; i++)
65	Runtime	strcpy(subject, "Programming") - 11 chars into char subject[8], buffer overflow	char subject[15]; strcpy(subject, "Programming");
84	Runtime	fees[5] = 100000 - index 5 out of bounds for fees[5] (valid: 0-4)	Remove fees[5] or declare fees[6]
123	Runtime	inventory["one"] = 100 - string as array index is invalid/crash	inventory[0] = 100;
125	Runtime	inventory[2.5] = 200 - float as array index is invalid	inventory[2] = 200;
131	Runtime	for(i=0; i<=5; i++) on inventory[5] - out of bounds at index 5	for(int i=0; i<5; i++)
145	Runtime	strcpy(course, "Computer") - 8 chars into char course[6], buffer overflow	char course[10];
150	Runtime	for(i=0; i<=5; i++) on numbers[5] - out of bounds at index 5	for(int i=0; i<5; i++)

Step 7: Logical Errors

Line	Error Type	Description	Correction
19	Logical	average = total / 5 - integer division; stores 83 instead of 83.0 in float	average = total / 5.0;
38	Logical	Finding highest: if(marks[i] > highest) correct here	No correction needed - this is correct
45	Logical	Finding lowest: if(marks[i] > lowest) - wrong condition; replaces lowest with LARGER values, finds highest not lowest	if(marks[i] < lowest)
91	Logical	if(stock[2] = 30) - assignment not comparison; always true	if(stock[2] == 30)

Step 8: Check Arrays Carefully

Line	Error Type	Description	Correction
60	Array	Loop i<=5 on attendance[5] - out of bounds	Change to i<5

Line	Error Type	Description	Correction
65	Array	char subject[8] too small for 11-char string "Programming"	char subject[15]
84	Array	fees[5] = 100000 - out of bounds for fees[5]	Remove or declare fees[6]
95	Array	backup = stock - array assignment not allowed in C	Use memcpy()
123	Array	inventory["one"] - string index invalid	Use integer index
125	Array	inventory[2.5] - float index invalid	Use integer index
131	Array	Loop i<=5 on inventory[5] - out of bounds	Change to i<5
136	Array	college[5] has no null terminator; printf("%s") may print garbage	Add college[5] = '\0'; or declare college[6]
139	Array	count = inventory - array assignment not allowed	Use memcpy()
145	Array	char course[6] too small for 8-char string "Computer"	char course[10]
150	Array	Loop i<=5 on numbers[5] - out of bounds	Change to i<5

Step 9: Check Functions Carefully

None found in this program.

Step 10: Check Structures

None found in this program.

Debugging Checklist - Program 3

☒ Syntax Errors

☐ Undeclared Variables

☒ Type Mismatches

☐ Function Call Errors

☐ Scope Errors

☐ Shadowing Issues

☐ Linker Errors

☒ Array Bounds Errors

☐ Pointer Errors

☐ File Handling Errors

☒ Runtime Errors

☒ Logical Errors

☒ Wrong Formulas

☒ Wrong Conditions

PROGRAM 4: Sales Report System

Step 1: Read the Entire Program

What is this program trying to do?

This program generates a sales report. It calculates total and average sales, finds the highest sale, manages monthly sales, handles product/category names, stock, orders, revenue, commission, city, branch, employees, department, expenses, ratings, and feedback data.

Identified Elements:

- Variables: totalSales, averageSales, highestSale, totalOrders
- Arrays: sales[5], monthlySales[5], product[8], category[15], stock[5], orders[5], revenue[5], backupRevenue[5], commission[5], city[10], branch[5], employees[5], department[6], expenses[5], ratings[5], feedback[5]
- Functions: main() only
- Loops: multiple for loops
- Structures: None
- File Operations: None
- Pointers: None

Step 2: Syntax Errors

Line	Error Type	Description	Correction
175	Syntax	Missing semicolon: printf("Report Generated Successfully\n")	printf("Report Generated Successfully\n");

Step 3: Semantic Errors

Line	Error Type	Description	Correction
21	Semantic	printf("Average Sales = %d", averageSales) but averageSales is float - wrong format specifier	printf("Average Sales = %f\n", averageSales);
59	Semantic	category = "Electronics" - cannot assign string to char array after declaration	strcpy(category, "Electronics");
64	Semantic	stock["first"] = 100 - string used as array index, invalid	stock[0] = 100;
66	Semantic	stock[2.7] = 200 - float used as array index, invalid	stock[2] = 200;
74	Semantic	scanf("%d", orders[i]) - missing & for address	scanf("%d", &orders[i]);
86	Semantic	if(revenue[2] = 3000) - assignment instead of comparison	if(revenue[2] == 3000)
92	Semantic	backupRevenue = revenue - cannot copy array by direct assignment in C	Use memcpy(backupRevenue, revenue, sizeof(revenue));
99	Semantic	printf("%d", commission[0]) but commission[0] is float - wrong format specifier	printf("%f\n", commission[0]);

Line	Error Type	Description	Correction
104	Semantic	city = "Jaipur" - cannot assign string to char array after declaration	strcpy(city, "Jaipur");
158	Semantic	expenses = sales - cannot copy array by direct assignment in C	Use memcpy(expenses, sales, sizeof(sales));
165	Semantic	printf("%s", feedback[0]) but feedback[0] is int - wrong format specifier	printf("%d\n", feedback[0]);

Step 4: Scope Errors

None found in this program.

Step 5: Linker Errors

None found in this program.

Step 6: Runtime Errors

Line	Error Type	Description	Correction
41	Runtime	monthlySales[5] = 250 - index 5 out of bounds for monthlySales[5] (valid: 0-4)	Remove or declare monthlySales[6]
45	Runtime	for(i=0; i<=5; i++) on monthlySales[5] - accesses monthlySales[5] out of bounds	for(int i=0; i<5; i++)
51	Runtime	strcpy(product, "LaptopComputer") - 14 chars into char product[8], buffer overflow	char product[20]; strcpy(product, "LaptopComputer");
64	Runtime	stock["first"] = 100 - string as index causes crash	stock[0] = 100;
66	Runtime	stock[2.7] = 200 - float as index is invalid	stock[2] = 200;
113	Runtime	branch[5] has 4 chars but no null terminator; printf("%s") may print garbage	Add branch[4] = '\0'; or declare branch[5] and null terminate
119	Runtime	for(i=0; i<=5; i++) on employees[5] - out of bounds at employees[5]	for(int i=0; i<5; i++)
127	Runtime	strcpy(department, "Marketing") - 9 chars into char department[6], buffer overflow	char department[12]; strcpy(department, "Marketing");
162	Runtime	ratings[-1] = 5 - negative index is out of bounds and undefined behavior	ratings[0] = 5; (use valid index)

Step 7: Logical Errors

Line	Error Type	Description	Correction
19	Logical	averageSales = totalSales / 5 - integer division; decimal part lost	averageSales = totalSales / 5.0;
27	Logical	Finding highest: if(sales[i] < highestSale) replaces highest with smaller value - finds lowest, not highest	if(sales[i] > highestSale) to find actual highest

Line	Error Type	Description	Correction
86	Logical	if(revenue[2] = 3000) assigns 3000 instead of comparing - always true	if(revenue[2] == 3000)

Step 8: Check Arrays Carefully

Line	Error Type	Description	Correction
41	Array	monthlySales[5] = 250 - out of bounds write	Remove or use monthlySales[6]
45	Array	Loop i<=5 on monthlySales[5] - out of bounds	Change to i<5
51	Array	char product[8] too small for 14-char "LaptopComputer"	char product[20]
64	Array	stock["first"] - string index invalid	Use integer index stock[0]
66	Array	stock[2.7] - float index invalid	Use integer index stock[2]
92	Array	backupRevenue = revenue - array assignment not allowed	Use memcpy()
119	Array	Loop i<=5 on employees[5] - out of bounds	Change to i<5
127	Array	char department[6] too small for 9-char "Marketing"	char department[12]
158	Array	expenses = sales - array assignment not allowed	Use memcpy()
162	Array	ratings[-1] - negative index, undefined behavior	Use valid index ratings[0]

Step 9: Check Functions Carefully

None found in this program.

Step 10: Check Structures

None found in this program.

Debugging Checklist - Program 4

- [YES] Syntax Errors
- [] Undeclared Variables
- [YES] Type Mismatches
- [] Function Call Errors
- [] Scope Errors
- [] Shadowing Issues
- [] Linker Errors
- [YES] Array Bounds Errors
- [] Pointer Errors
- [] File Handling Errors

[YES] Runtime Errors

[YES] Logical Errors

[YES] Wrong Formulas

[YES] Wrong Conditions

PROGRAM 5: Student Performance Analysis

Step 1: Read the Entire Program

What is this program trying to do?

This program analyzes student performance. It calculates total marks, average, highest and lowest from a 10-element array, tracks attendance, manages fees, result input, inventory, salary, city, branch, stock, course, ratings, feedback, scores, and a temp array.

Identified Elements:

- Variables: total, average, highest, lowest, present, totalResult
- Arrays: marks[10], attendance[10], subject[10], department[20], fees[5], result[5], inventory[5], backup[5], salary[5], city[10], branch[5], stock[5], course[8], ratings[5], feedback[5], scores[5], temp[5]
- Functions: main() only
- Loops: for loops for traversal, filling, and input
- Structures: None
- File Operations: None
- Pointers: None

Step 2: Syntax Errors

None found in this program.

Step 3: Semantic Errors

Line	Error Type	Description	Correction
22	Semantic	printf("Average = %d", average) but average is float - wrong format specifier	printf("Average = %f\n", average);
57	Semantic	department = "InformationTechnology" - cannot assign string to char array after declaration	strcpy(department, "InformationTechnology");
62	Semantic	fees["two"] = 50000 - string used as array index, invalid	fees[1] = 50000;
64	Semantic	fees[4.5] = 70000 - float used as array index, invalid	fees[4] = 70000;
73	Semantic	scanf("%d", result[i]) - missing & for address	scanf("%d", &result[i]);
86	Semantic	if(inventory[2] = 30) - assignment instead of comparison	if(inventory[2] == 30)
92	Semantic	backup = inventory - cannot copy array by direct assignment in C	Use memcpy(backup, inventory, sizeof(inventory));
99	Semantic	printf("%d", salary[0]) but salary[0] is float - wrong format specifier	printf("%f\n", salary[0]);
104	Semantic	city = "Jaipur" - cannot assign string to char array after declaration	strcpy(city, "Jaipur");
133	Semantic	printf("%s", feedback[0]) but feedback[0] is int - wrong format specifier	printf("%d\n", feedback[0]);

Line	Error Type	Description	Correction
137	Semantic	scores = marks - cannot copy array by direct assignment in C	Use memcpy(scores, marks, sizeof(scores));

Step 4: Scope Errors

None found in this program.

Step 5: Linker Errors

None found in this program.

Step 6: Runtime Errors

Line	Error Type	Description	Correction
51	Runtime	attendance[10] = 0 - index 10 out of bounds for attendance[10] (valid: 0-9)	Remove or declare attendance[11]
55	Runtime	for(i=0; i<=10; i++) on attendance[10] - accesses attendance[10] out of bounds	for(int i=0; i<10; i++)
61	Runtime	strcpy(subject, "DataStructures") - 14 chars into char subject[10], buffer overflow	char subject[20]; strcpy(subject, "DataStructures");
57	Runtime	strcpy(department, "InformationTechnology") - 21 chars into char department[20], buffer overflow	char department[25]; strcpy(department, "InformationTechnology");
62	Runtime	fees["two"] = 50000 - string as index causes crash	fees[1] = 50000;
64	Runtime	fees[4.5] = 70000 - float as index is invalid	fees[4] = 70000;
110	Runtime	branch[5] declared with 4 chars and no null terminator; printf("%s") may print garbage	Add branch[4] = '\0'; or declare branch[5] properly
115	Runtime	for(i=0; i<=5; i++) on stock[5] - accesses stock[5] out of bounds	for(int i=0; i<5; i++)
121	Runtime	strcpy(course, "Programming") - 11 chars into char course[8], buffer overflow	char course[15]; strcpy(course, "Programming");
125	Runtime	ratings[-1] = 10 - negative array index is undefined behavior / crash	ratings[0] = 10;
143	Runtime	printf("%d", temp[5]) - index 5 out of bounds for temp[5] (valid: 0-4)	printf("%d\n", temp[4]);

Step 7: Logical Errors

Line	Error Type	Description	Correction
20	Logical	average = total / 10 - integer division; decimal part lost before storing in float	average = total / 10.0;

Line	Error Type	Description	Correction
29	Logical	Finding highest: if(marks[i] < highest) replaces highest with smaller value - finds lowest not highest	if(marks[i] > highest)
39	Logical	Finding lowest: if(marks[i] > lowest) replaces lowest with larger value - finds highest not lowest	if(marks[i] < lowest)
86	Logical	if(inventory[2] = 30) assigns 30 to inventory[2] instead of comparing - always true	if(inventory[2] == 30)

Step 8: Check Arrays Carefully

Line	Error Type	Description	Correction
51	Array	attendance[10] = 0 - out of bounds for attendance[10]	Remove or use attendance[11]
55	Array	Loop i<=10 on attendance[10] - out of bounds	Change to i<10
57	Array	char department[20] too small for 21-char "InformationTechnology"	char department[25]
61	Array	char subject[10] too small for 14-char "DataStructures"	char subject[20]
62	Array	fees["two"] - string index invalid	Use integer index fees[1]
64	Array	fees[4.5] - float index invalid	Use integer index fees[4]
92	Array	backup = inventory - array assignment not allowed in C	Use memcpy()
115	Array	Loop i<=5 on stock[5] - out of bounds	Change to i<5
121	Array	char course[8] too small for 11-char "Programming"	char course[15]
125	Array	ratings[-1] - negative index undefined behavior	Use valid index ratings[0]
137	Array	scores = marks - array assignment not allowed in C	Use memcpy()
143	Array	temp[5] access - out of bounds for temp[5] (valid: 0-4)	Use temp[4]

Step 9: Check Functions Carefully

None found in this program.

Step 10: Check Structures

None found in this program.

Debugging Checklist - Program 5

- ☐ Syntax Errors
- ☐ Undeclared Variables
- ☒ Type Mismatches
- ☐ Function Call Errors
- ☐ Scope Errors
- ☐ Shadowing Issues
- ☐ Linker Errors
- ☒ Array Bounds Errors
- ☐ Pointer Errors
- ☐ File Handling Errors
- ☒ Runtime Errors
- ☒ Logical Errors
- ☒ Wrong Formulas
- ☒ Wrong Conditions